

```

string prevData = "temp";

while (true)
{
    try
    {
        /* If the current data is not equal
        to the
        * old data then the TextBlock for the
        Rfid
        * is updated */
        if (!prevData.Equals(data))
        {
            prevData = data;
            rfId.Dispatcher.Invoke((Action)() =>
            {
                rfId.Text = prevData;
            });
        }
        catch (Exception ex)
        {
            Console.WriteLine("Error: " + ex);
        }
    }
}

```

Now, we need to add a reference of AttendanceManagementLibrary in the client-side program and then write the event handler for Search button. Inside the event handler, create an instance of ServiceClient, which is inside ServiceReference1. Then, open the service-client instance so that we can access the server functionality. We get data from the database through the server-side based on EmpId set by the user in the UI and create our own data-model object that we assign to DataContext property of the concerned textblock.

At the end, we need to write the method for Register button. When Register button is clicked in the UI, we take all details from the UI and construct our client-data model that we pass to the server using CreateEmpDetails method. After data is successfully entered into the database, 'Registered!!!' text pops up in the UI.

Testing the project

1. Connect an RFID reader to the system where SQL Server 2008 and its related tools are installed.

EmpId	Position	LoginTime	LogoutTime	LoginCount
IC1	efy_chennai	08-01-2015 14:00	11-01-2015 14:00	2
IC2	efy_bangalore	08-01-2015 14:00	11-01-2015 14:00	2
IC3	efy_kolkata	08-01-2015 14:00	11-01-2015 14:00	2

Fig. 25: Saved data of current employees



Fig. 26: RFID reader



Fig. 27: RFID tags

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

2. Now, open Visual Studio 2010, go to AttendanceServerSide solution and update DataSource (by taking the name from SQL Server Management Studio) of App.config files present in AttendanceRegistrationSystem and AttendanceRegisterServer projects to point to your own DataSource name where the database is present.

3. Build AttendanceRegistrationSystem solution.

4. Go to the respective debug or release folders of AttendanceRegisterServer project in Attend-

anceServerSide and AttendanceManagementSystemClient project in AttendanceClientSide applications and copy both folders in the system, which is already attached with the RFID reader.

5. Now, first run AttendanceRegisterServer.exe and let it continue to run in the background and then run AttendanceManagementSystemClient.exe, both in administrator mode. The program output screen of AttendanceManagementSystemClient program is shown in Fig. 1 in part 1 of the article.

Then, you can insert or retrieve data through this client program. To insert data, place an RFID tag near the RFID reader; the client program will read the tag number and display the same on the screen automatically. Enter the name, address and serial port number, and then click Register. After the data is successfully entered into the database, the 'Registered!!!' message pops-up in the UI. User (employee) IDs, like IC0, IC1 and IC2, are generated by the program automatically. The admin can view details of a particular employee by clicking on Employee Data, followed by entering the ID number.

6. All saved data can be viewed in the database tables by clicking View->Server Explorer->Tables. Right-click on CurrEmpDetail from Server Explorer to view the saved data as shown in Fig. 25.

LoginTime is registered whenever a tag is read. When the same tag is read again by the RFID reader, the employee's LogoutTime will be updated in the database (as per our logic). The RFID reader and RFID tags used in this project are shown in Figs 26 and 27, respectively. ●

Concluded



Sunanda Chatterjee is B.Tech (IT) from JIS College of Engineering, Kalyani. She is working as product specialist (technical), Cognizant Technology Solutions, Kolkata